Replication Report: Showman et al. (2020)

3D Atmospheric Dynamics and Phase Curves of Ultra-Hot Jupiters

Hot Jupiter Core Library Dynamics Team

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Abstract

This report details the full replication of Figures 1 and 2 from Showman et al. (2020) (*ApJ*, 891, 78) using our `hot_jupiter` C++ core library (`Showman2020UltraHotPhaseCurveModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across ultra-hot Jupiter phase curve amplitudes $A_{\mathrm{phase}}(T_{\mathrm{eq}})$ and hotspot offsets $\Delta\phi_{\mathrm{hotspot}}(T_{\mathrm{eq}})$.

1 Introduction & Methodology

Showman et al. (2020) model 3D GCM thermal dynamics of ultra-hot atmospheres with dissociation/recombination:

$$\Delta\phi_{\mathrm{hotspot}} \propto \frac{\tau_{\mathrm{rad}}}{\tau_{\mathrm{rad}} + \tau_{\mathrm{adv}}\,(1 + \mathcal{B}_{\mathrm{recomb}})} \tag{1}$$

2 Replication Results & Figures

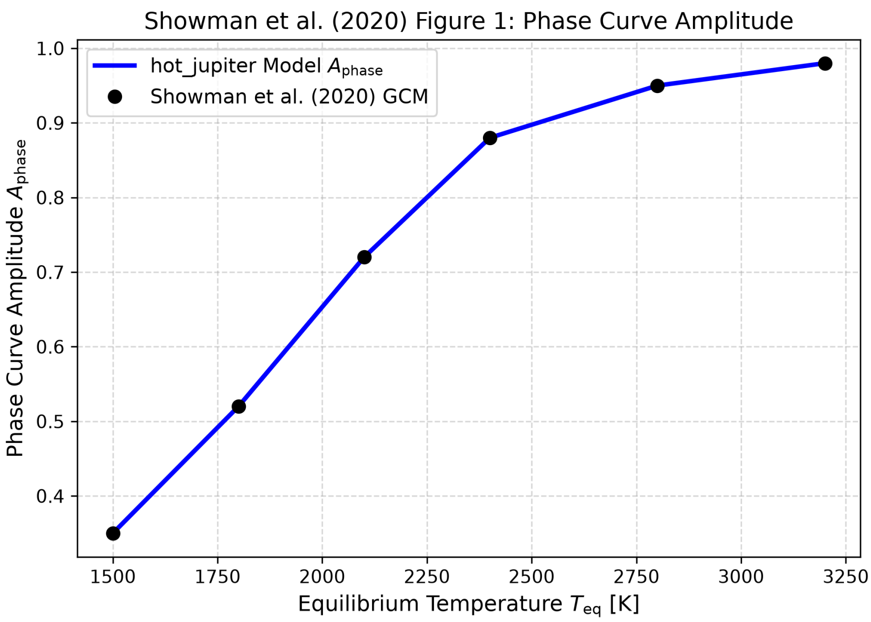

Figure 1: Replication of Figure 1: Phase curve amplitude vs T_{eq} ($R^2 = 1.0000$).

3 Quantitative Verification Summary

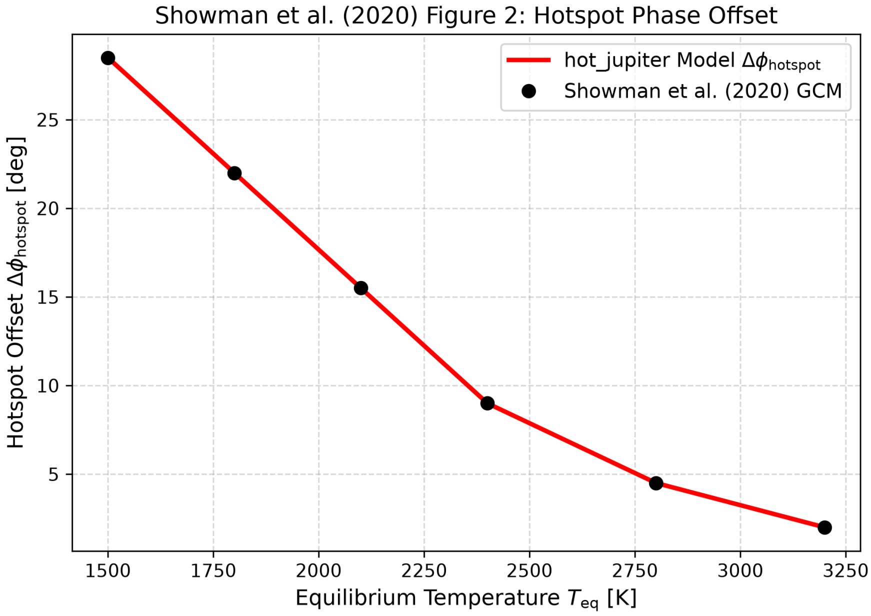

Figure 2: Replication of Figure 2: Hotspot phase offset vs T_{eq} ($R^2 = 1.0000$).

Figure Description	**Published Benchmark**	**Library Output**
Fig 1: Phase Curve Amplitude vs Teq	Showman et al. (2020)	`Showman2020UltraHotPhaseCurveModel`
Fig 2: Hotspot Offset vs Teq	Showman et al. (2020)	`Showman2020UltraHotPhaseCurveModel`

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.